

Clarke Needles

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EDUCATION

Queen's University

Bachelor of Applied Science, Computer Engineering (3.5 GPA)

Kingston, ON

2023 – Present

- **Interest:** Embedded Software Development, Embedded Firmware, Embedded Systems.
- **Related Coursework:** Microprocessor Interfacing and Embedded Systems, Operating Systems, Computer Architecture, Digital Systems.

PROFESSIONAL EXPERIENCE

FUJIFILM VisualSonics

Software Engineering Intern

Toronto, ON

May 2025 – August 2025

- Developed C/C++ system applications using WinAPI, ATL, and COM.
- Designed and implemented a TreeView file navigator panel using the MVP pattern.
- Improved service team productivity through new file navigator feature.
- Contributed ~16,000+ lines of code across the main code base, reviewed and approved by senior developers.

Quality Assurance Engineering Intern

May 2024 – August 2024

- Automated 4 system validation procedures and 300+ test cases (~12,000+ lines of code) in C#.
- Personally increased overall system test coverage by 8%.
- Collaborated in a large agile development team, contributed to large scale production-grade codebase.

TECHNICAL EXPERIENCE

Embedded Peripheral Playground — STM32 + FreeRTOS

- STM32-based platform running a real-time operating system (FreeRTOS).
- Hardware interfaces including UART, SPI, I²C sensors, CAN bus communication.
- Implemented interrupt-driven drivers and real-time task scheduling for reliable data acquisition and control.
- Gained hands-on experience with embedded C, hardware abstraction layers, debugging, and peripheral interfacing.

Queen's Aerospace Design Team — Autonomy Software Developer

- Contributing to the design and implementation of embedded software development using C++ for UAV guidance, navigation, and flight control systems.
- Developing ROS2 software to interface with hardware to collect, process, and share data across the UAV system.

Queen's High Performance Computing Club — General Member

- Hands-on experience with Rocky Linux in a high-performance computing environment.
- Collaborating with team members on large scale simulations, benchmarking, and resource management for HPC clusters. Utilizing parallel programming models for parallel and distributed work loads.

Embedded Systems Lab — Nios II Assembly (Altera DE0)

- Developed and tested assembly routines for the Nios II soft-core processor on an Altera DE0 board.
- Implemented interrupt service routines, I/O handling, and bit-level control of hardware peripherals.
- Gained practical experience with low-level firmware design, memory mapped I/O, and real-time processing concepts.

Audio Gain Amplifier — BJT Design

- Built and tested a BJT-based amplifier with a regulated AC–DC power supply.
- Analyzed circuit performance, gain characteristics, and power efficiency.

Parcel Shield — Arduino + MQTT + Flutter

- Created a package security system using Arduino, external sensors, and an MQTT broker.
- Designed a Flutter-based web app for remote monitoring and control.

SKILLS

Programming: C, C++, Python, Nios II Assembly, C#, Java, Git

Embedded Systems: STM32, FreeRTOS, UART, SPI, I²C, CAN, RTOS concepts

Testing & QA: Unit testing, integration testing, automation, debugging

Tools & Platforms: Visual Studio, ROS2, Arduino IDE, WinAPI, Jira, Jenkins, SVN